Recipients: Ali Kumail (Full Stack) · Ali Hafidz (UX/UI) · V4 replaces all previous versions

WHAT IS IT

Doc C defines the canonical data structure for all observations and measurements in NuBred. It is the reference for database schema design, UI form building, the AI extraction pipeline, and the decision logic engine. Every piece of data collected in NuBred maps to this document.

V4 major changes from V3: (1) 8 parameter families — F8 Agronomic Operations added. (2) Three levels of parameters — Universal, Species-standard, Project-custom. (3) Two protocol types — Observational (Trial/Pilot) and Commercial (Launch/Scale). (4) Input type Scale is now configurable min–max, not hardcoded 1–5. (5) Distribution added as input type for calibre and histogram data. (6) Four frequency types including Event-based and Phase-gate. (7) Full kiwi species library for MVP.

THE 8 PARAMETER FAMILIES

All parameters in NuBred belong to one of 8 families. Families are fixed — they never change. What changes per species and per project is which parameters within each family are active, with what thresholds, and with what input types.

Family	Name	What it measures	Protocol type	Royalty input
F1	Plant	Vigour, growth habit, phenology, trunk/stem diameter, plant health, branching density	Observational	No
F2	Fruit Quality	Weight, size, shape, colour, Brix, acidity, firmness, juiciness, bloom, luminosity, organoleptic	Observational	Quality Spec input
F3	Production	Kg/plant or kg/ha, pack-out %, calibre distribution, quality grade, commercial vs reject	Observational + Commercial	**Yes — delta EBITDA**
F4	Phytosanitary	Pest/disease identification, affectation %, severity scale, cracking, physiological disorders	Observational	Eliminatory input
F5	Post-Harvest	Storage duration, % damaged/rotten/pitted, freshness, bloom retention, weight loss, shelf life	Observational	Quality Spec input
F6	Climate	Temperature, humidity, wind, radiation, precipitation, light intensity, incidents	Contextual	Context only
F7	Commercial	Production forecast (4 measurements), harvest actuals, milestone hectares, royalty flows	Commercial only	**Yes — direct**
F8	Agronomic Operations	Irrigation (volumes, intervals, water quality), fertilization, pest/disease treatments with products and doses, pruning, soil management	Operational log	**Cost side — delta EBITDA**

F8 Agronomic Operations is new in V4. It covers irrigation, fertilization, pest/disease treatments, pruning, and soil management. This is not an observation — it is an operational log. It is the context that explains why varieties perform differently across sites, and it is the cost input for the delta EBITDA calculation in the Advanced Protocol.

THREE LEVELS OF PARAMETERS

Not all parameters are equal. NuBred organizes parameters into three levels based on their scope and who controls them. This is what allows NuBred to be species-flexible while maintaining governance integrity across all projects.

Level	Name	Definition	Who controls it
Level 1	Universal parameters	Valid for all species in all phases. Always included in every protocol. Cannot be removed. Examples: temperature (F6), production weight (F3), pack-out % (F3).	NuBred — fixed. No one can remove these.
Level 2	Species-standard parameters	Defined in the NuBred Parameter Registry for a specific species. Proposed automatically when VM selects species. VM can raise thresholds but cannot remove mandatory species parameters. Examples: Brix for strawberry and blueberry, Bloom for blueberry and grape, cracking for cherry.	NuBred proposes per species. VM can raise thresholds, never lower. Cannot remove mandatory ones.
Level 3	Project-custom parameters	Added by VM for specific project requirements — market-specific, client-specific, or research-specific. Examples: proline content for plum in a specific market, luminosity L* for premium strawberry, mechanical harvest resistance. Not in standard Registry.	VM defines freely. Stored as project-specific parameters. Do not affect other projects.

Example: Proline content is a Level 3 parameter — it matters for plum or blueberry in specific markets (antioxidant content claims) but is not in the standard Registry. A VM working on blueberry for a health-food retailer adds it as a project-custom parameter. Another VM working on the same species for a supermarket does not need it. Both projects use the same NuBred schema.

TWO PROTOCOL TYPES — OBSERVATIONAL vs COMMERCIAL

This is the most important architectural distinction in V4. Protocols are not a single monolithic schema — they split into two fundamentally different types depending on the project phase. Confusing them creates the wrong data model.

Protocol type	Active phases	Primary objective	Key output
Observational	Trial, Pilot	Scientific evaluation of agronomic, qualitative, and phytosanitary performance. Supports the Promote / Repeat / Discard decision for each genotype.	Trial report (like ALSIA format). Phase gate decision. Weighted score per genotype.

Commercial	Launch, Scale	Production tracking and royalty calculation. Observational parameters reduced to Quality Specification minimums. F7 Commercial active with 4-measurement forecast cycle.	Production forecast. Harvest Actual declaration. Royalty calculation (Standard or Advanced/Delta EBITDA).

The Observational protocol produces a scientific trial report — like the ALSIA Final Report commissioned by SanLucar. The Commercial protocol produces a royalty calculation. They use different parameter families, different frequencies, different outputs. A project moving from Pilot to Launch does not continue the same protocol — it switches protocol type entirely.

OBSERVATIONAL PROTOCOL — PARAMETER SCHEMA

The following parameters apply to Trial and Pilot phases. They are the observational layer — the scientific data that feeds the Promote/Repeat/Discard decision. Species-specific examples use strawberry and kiwi. Full species libraries are maintained in the NuBred Parameter Registry.

Parameter	F	How to measure	Input type	Frequency	Threshold type	Decision role
Plant Vigour	F1	Visual scale or trunk/stem circumference	Scale (custom)	Bi-weekly / Monthly	UPOV qualitative	Weighted — species profile
Phenology (BBCH)	F1	BBCH scale — visual + photo	Option (BBCH)	Weekly (active season)	Qualitative absolute	Weighted — weight 10%
Growth Habit	F1	UPOV descriptor — visual classification	Option (UPOV)	1× / season	UPOV qualitative	Eliminatory L2 — must match training system
Brix (°)	F2	Refractometer on juice — min 5 fruits	Number	1× at harvest or monthly	Quantitative absolute	Eliminatory L2 — species minimum
Fruit Firmness	F2	Penetrometer (N or kg/cm²) or sensory scale	Number or Scale	1× at harvest	Quantitative relative	Weighted — weight 20%
Colour	F2	Colorimeter (L* a* b*) or visual scale	Number or Scale	1× at harvest	UPOV qualitative	Weighted — weight 10%
Average Fruit Weight (g)	F2	Mean weight of 20 random fruits — precision scale	Number	1× at harvest	Quantitative absolute	Weighted — weight 15%
Bloom (species-specific)	F2	Visual scale — waxy coating on fruit surface	Scale 1–3	1× at harvest	UPOV qualitative	Weighted — blueberry/grape/plum only
Organoleptic (Aspect + Flavour)	F2	Panel tasting — trained evaluators or consumer panel	Scale 1–5	3× per season	UPOV qualitative	Weighted — commercial quality
Commercial Production (kg/plant or kg/ha)	F3	Weigh all harvested commercial fruit per plot — precision scale	Number	Per harvest event	Quantitative relative vs benchmark	**Delta EBITDA input — direct**
Pack-out % (1ª cat vs reject)	F3	Commercial grade weight / total harvested weight	Number (%)	Per harvest event	Quantitative relative vs benchmark	Delta EBITDA input
Calibre distribution (%)	F3	Calibration rings — 20 fruits per session, % per size class	Distribution	Per harvest event or monthly	Quantitative absolute	Quality Spec input — size classes
Disease severity (per pathogen)	F4	% affected + severity scale 1–5 per pathogen	% + Scale 1–5	Monthly	Threshold absolute	**Eliminatory L2 — species threshold**

		identified in species library				
Pest affectation (per pest)	**F4**	% affected plants + severity scale 1–5	% + Scale 1–5	Monthly	Threshold absolute	Eliminatory L2 — species threshold
Shelf Life (days at temp)	**F5**	Evaluate at defined intervals: 1d, 3-4d, 7d, 14d, 21d post-harvest at ambient and cold	Number (days)	3× per season	Quantitative relative vs benchmark	Quality Spec input
Post-harvest quality (freshness, % damaged)	**F5**	Visual scale freshness (1–3), % damaged/rotten, fit for sale (Y/N)	Scale + % + Boolean	At each shelf life interval	Quantitative absolute + Boolean	Quality Spec gate
Temperature (°C)	**F6**	On-site weather station — daily min/mean/max	Number	Daily	Context only	No threshold — contextual record
Light intensity (klux)	**F6**	Lux sensor — max and mean per day. Relevant for greenhouse trials.	Number	Daily	Context only	No threshold — contextual record

Note on calibre distribution: this parameter uses the Distribution input type — a set of percentage values across defined size classes that must sum to 100%. It cannot be represented as a single scalar value. Ali Kumail: ObservationValue must support structured values for this parameter type, not only scalars.

COMMERCIAL PROTOCOL — F7 PARAMETER SCHEMA

The following parameters apply to Launch and Scale phases only. They replace the full observational schema — quality parameters are reduced to Quality Specification minimums. The focus shifts entirely to production forecasting and royalty calculation.

Parameter	F	How to measure	Input type	Timing	Role
Fruit Set Count	**F7**	Count fruitlets per sample area — extrapolate to kg/ha	Number	Measurement 1 — early season	Initial production estimate. Widest uncertainty range. Triggers logistics planning.
Growing Stage Diameter	**F7**	Average diameter of 50 sample fruits — calipers	Number (mm)	Measurement 2 — mid season	Refined estimate. Growth curve applied. Uncertainty narrows significantly.
Ripening Stage Weight	**F7**	Final field weight check on representative sample	Number (g/fruit)	Measurement 3 — pre-harvest	Final pre-harvest forecast. Basis for provisional royalty invoice and logistics.

Harvest Actual	**F7**	Total weighed harvest per plot or per grower — packhouse or field scale	Number (kg/ha)	Measurement 4 — post-harvest	DEFINITIVE figure. Final royalty calculation basis. Compared vs Measurement 3 for forecast accuracy tracking.

The 4-measurement cycle is mandatory from the Pilot phase onward. Measurement 4 (Harvest Actual) is the definitive figure that triggers royalty calculation. All previous measurements are forecasts — they improve logistics planning and allow early royalty invoicing but are superseded by Measurement 4.

F8 — AGRONOMIC OPERATIONS

F8 is a new family in V4. It is not an observation — it is an operational log of all agronomic interventions during the trial. It is mandatory for any trial that will produce a certified report (GLP-compliant, accredited institution format). Without it, inter-variety comparisons are not valid because unknown agronomic differences between plots cannot be excluded as confounding factors.

Operation type	Key fields	Examples	Why it matters for NuBred
Irrigation	Date, volume (m^3/ha), interval (days), water pH, EC (µS/cm)	100 m^3/ha every 2-3 days from April to May	Contextual interpretation of production and quality results. Cost input for delta EBITDA.
Fertilization / Foliar treatment	Date, product name, active substance, dose (g or ml per hl), application method	Switch (Ciprodinil 37.5% + Fludioxonil 25%) 100g per hl — 04/02/2026	Explains disease resistance differences between plots. Required for GLP/accredited trial report.
Pruning / Training	Date, type (pruning / defoliation / topping), intensity	Defoliation of basal leaves 27/02/2026	Affects fruit quality and production timing. Essential context for inter-variety comparison.
Soil management	Date, type (hoeing, weeding, mulching)	Manual hoeing inter-row 28/11/2025 and 10/02/2026	Documents agronomic standardization across plots — required for valid variety comparison.

F8 is also the cost side of the delta EBITDA calculation. In the Advanced Protocol, royalties are calculated on the economic delta between the protected variety and the benchmark. That delta depends not only on production and quality (revenue side) but also on the cost of inputs and operations (cost side). Without F8, delta EBITDA cannot be calculated correctly.

INPUT TYPES — COMPLETE DEFINITION

Every parameter has exactly one input type. The input type determines the UI component shown to the Field Observer, the validation rules applied to the value, and how the data is stored in ObservationValue. Ali Kumail: ObservationValue.value must be typed at the application layer based on the parameter's input type — not stored as raw text.

Input type	Definition	Examples in protocols

Number	Single numeric value with unit.	Brix 11.2°, Weight 18.4g, Temperature 22.6°C, Production 4.2 kg/plant
Scale (min–max)	Ordinal scale with defined min and max. Min and max are configurable per parameter — not hardcoded to 1–5. Polarity (higher=better or lower=better) is defined in the Registry.	Vigour 1–3 (1=small, 3=large), Firmness 1–5 (1=soft, 5=firm), Bloom 1–3 (1=low, 3=high)
Option (list)	Predefined list of values. List is defined in the Parameter Registry per species.	BBCH stage (00, 10, 51, 61...), Growth habit (erect, semi-erect, spreading), Insolation (sunny, partly cloudy, cloudy, rain)
Distribution	Set of values across defined categories that must sum to 100%. Used when a parameter is inherently a histogram rather than a single value.	Calibre distribution: <12mm 5%, 12-14mm 18%, 14-18mm 42%, 18-20mm 28%, >20mm 7%
Date	Calendar date of an event. Defaults to today but editable.	First flowering date, first harvest date, last harvest date
Boolean (Y/N)	Binary — Yes or No. Always eliminatory when used as a threshold.	Fit for sale after shelf life: Yes/No. Quarantine pathogen detected: Yes/No
Photo	Camera capture in mobile app. Linked to observation session and genotype. Stored in Cloudinary/Hetzner.	Field documentation, disease symptoms, fruit appearance at harvest and shelf life intervals
Text	Free text. Used only for notes, pest/disease identification where no predefined list exists, and Field Observer comments.	Incident notes, unusual observations, agronomic anomalies

Scale min–max replaces Scale 1–5 from previous versions. The previous hardcoded 1–5 was wrong for blueberry protocols (which use 1–3 extensively) and wrong for any species where a 1–10 or 1–7 scale is standard. The min, max, and polarity (higher=better or lower=better) are now configurable per parameter in the Registry.

FREQUENCY TYPES

Every parameter has a defined frequency. In V4, four frequency types are available. Calendar-fixed frequencies were the only type in previous versions. Event-based and Phase-gate are new and critical for protocols like the ALSIA strawberry report where quality measurements happen per harvest event, not on a fixed calendar.

Frequency type	Definition	Example
Calendar-fixed	Observation triggered on a fixed calendar schedule: daily, weekly, bi-weekly, monthly.	Climate data: daily. Plant vigour: bi-weekly during growth, monthly rest of season.
Event-based	Observation triggered by a biological or agronomic event, not a calendar date. The system generates the observation request when the event is logged.	Fruit quality: per harvest event. Phenology tracking: per BBCH stage transition. Shelf life: at 1d, 3-4d, 7d, 14d, 21d from harvest date.
Phase-gate	Observation required before phase advancement. Blocks the Promote decision until completed.	Quality Specification assessment: required before Trial→Pilot gate. Production forecast Measurement 3: required before Pilot→Launch gate.
On-demand	Observation triggered manually by VM or Field Observer when an anomaly or incident requires documentation.	Hail damage documentation. Unexpected disease outbreak. Physiological disorder on specific genotype.

KIWI SPECIES LIBRARY — MVP PARAMETER REGISTRY

This is the parameter Registry for Kiwi (Actinidia deliciosa and A. chinensis) — the species for the first real client MVP. All parameters below are Level 2 (species-standard) unless marked otherwise. The VM can add Level 3 custom parameters on top of these.

PSA (Pseudomonas syringae pv. actinidiae) is an L1 eliminatory parameter for yellow kiwi (Actinidia chinensis). Any confirmed PSA infection = immediate discard signal, no override possible. This is the most commercially devastating disease in kiwi production globally and must be a hard gate in NuBred.

Parameter	F	How to measure	Input type	Frequency	Threshold type	Kiwi-specific notes
Plant Vigour (cane growth)	**F1**	Visual or cane length measurement	Scale 1–5 or Number (cm)	Monthly	UPOV qualitative	Kiwi is vigorous — high vigor is not always positive for fruit quality
Phenology (BBCH)	**F1**	BBCH scale — bud burst to fruit maturity	Option (BBCH)	Weekly active season	Qualitative absolute	Harvest date predictability is key commercial parameter for kiwi
Dry Matter % (DM)	**F2**	Microwave or oven drying method — min 10 fruits	Number (%)	3× pre-harvest + at harvest	Quantitative absolute	**CRITICAL for kiwi — DM at harvest determines storability. Minimum DM varies by market (NZ: 6.2%, EU: 6.5%). Harvest gate parameter.**
Brix (°) at consumption	**F2**	Refractometer after 2-3 weeks ripening at ambient temperature	Number	3× per season	Quantitative absolute	Kiwi is measured at consumption Brix, not harvest Brix. Different from strawberry protocol.
Fruit Firmness (kg/0.5cm²)	**F2**	Penetrometer with 0.5cm² tip — min 10 fruits	Number (kg/0.5cm²)	At harvest + at consumption readiness	Quantitative absolute	Harvest firmness >4 kg/0.5cm² typically required. Firm kiwi = longer shelf life.
Average Fruit Weight (g)	**F2**	Mean weight of 20 random fruits	Number	At each harvest	Quantitative absolute	Key commercial parameter — market size requirements vary. >100g for premium EU market.
PSA susceptibility (Pseudomonas syringae pv. actinidiae)	**F4**	Visual symptoms + lab confirmation if suspected	Boolean + Text	Monthly	Boolean eliminatory	**ELIMINATORY L1 for yellow kiwi. Any confirmed PSA = immediate discard signal. No exception.**
Botrytis cinerea affectation %	**F4**	% affected fruit at harvest	% + Scale 1–5	At harvest	Threshold absolute	L2 eliminatory threshold. >10% commercial rejection risk.
Post-harvest shelf life (weeks)	**F5**	Evaluate at 4, 8, 12, 16 weeks in cold storage (0-1°C)	Number (weeks) + quality scale	3× per season	Quantitative relative vs benchmark	Kiwi shelf life is measured in weeks, not days. 4-month storability is the

						premium market benchmark.
Production (kg/plant)	**F3**	Total harvest weight per vine — precision scale	Number	Per harvest event	Quantitative relative vs benchmark	Kiwi measured per plant (vine), not per ha in Trial phase. Conversion to ha in Pilot.

Key kiwi-specific differences from other species: (1) DM% at harvest is the primary harvest gate parameter — more important than Brix in most markets. (2) Shelf life is measured in weeks, not days. (3) Production is measured per vine in Trial phase, converted to kg/ha in Pilot. (4) PSA is L1 eliminatory — absolute, no exceptions.

DATA SOURCE — THREE LEVELS

Every observation in NuBred has a data source. The source determines the confidence level and the icon shown in the UI. The data schema is identical across all three sources — only the source field changes. This is the architectural decision that makes NuBred Node integration require zero refactoring.

Source	Today	Data schema	Future
Manual	Field Observer enters data on mobile app according to active protocol.	ObservationSession { source: 'Manual', observerId, genotypeId, sessionDate, values[] }	Human NuBred Node. This is the MVP. Field Observer IS the Node.
Node	NuBred Node hardware sensor transmits data automatically via API.	ObservationSession { source: 'Node', nodeId, genotypeId, sessionDate, values[] }	Same schema as Manual. source field is the only difference. Zero refactoring required.
Vision	Computer vision system — camera + AI model — transmits structured data automatically.	ObservationSession { source: 'Vision', cameraId, modelVersion, genotypeId, sessionDate, values[] }	Future. Fruit counting, calibre measurement, colour, disease detection from images.

DECIDED — DO NOT REOPEN

- 8 parameter families — F8 Agronomic Operations added to the 7 existing families
- Three levels: Universal (all species), Species-standard (Registry per species), Project-custom (VM-defined)
- Two protocol types: Observational (Trial/Pilot) and Commercial (Launch/Scale) — they are different schemas
- Scale input type is configurable min–max — not hardcoded 1–5
- Distribution is a valid input type for calibre and histogram data — ObservationValue must support structured values
- Four frequency types: Calendar-fixed, Event-based, Phase-gate, On-demand
- DM% is the primary harvest gate parameter for kiwi — not Brix
- PSA is L1 eliminatory for yellow kiwi — absolute, no VM override
- Kiwi shelf life measured in weeks (4, 8, 12, 16 weeks) not days
- Three data sources: Manual / Node / Vision — identical schema, different source field only

OPEN — TEAM TO IMPLEMENT

- Parameter Registry database — species, parameter families, Level 1/2/3 structure, configurable Scale min/max and polarity
- Distribution input type handler — stores histogram as structured JSON, validates % sum to 100
- Event-based frequency trigger — observation request generated when harvest event is logged
- Phase-gate frequency — observation blocks Promote action until completed
- F8 AgronomicOperation entity in Prisma schema — date, operationType, product, dose, plot, notes
- Kiwi Registry population — all parameters from this section with 7 attributes each
- Source field on ObservationSession — Manual / Node / Vision with appropriate UI indicators

DO NOT

- Do not hardcode Scale as 1–5 anywhere in the codebase — always use configurable min/max
- Do not store calibre distribution as a single value — it is a structured histogram
- Do not apply Observational protocol parameters to Launch/Scale phases — they use Commercial protocol
- Do not allow VM to override PSA eliminatory signal for yellow kiwi — it is L1
- Do not build a different data schema for NuBred Node or Vision sources — source field only
- Do not skip F8 AgronomicOperation — it is required for certified trial reports and delta EBITDA